

C-LM: Chess Language Model



Goal

Finding out if chess can be played by a Language Model, i.e., a program that knows nothing about a board, pieces, or rules.

Dataset

- The data used for this project came from the *Lichess Database*, a free, CC0 resource that contains millions of chess games.
- There's a lot of extra data that can be had from these, but I only required the chess game moves themselves.

- e4 e5
- Nf3 Nc6
- Bc4 h6
- d4 exd4
- Nxd4 Nxd4
- Qxd4 d6
- e5 c5
- ...

Problems

- I decided to do this project because I wanted to better understand how LLMs work and what their limitations are.
- I thought the game of chess was a fun playground to figure that out on because it has very concrete rules and is fundamentally geometric, two things LLMs have no concept of.
- By design LLMs are obviously terrible at playing, which can be seen by trying to play vs the SOTA, so this was purely for fun and educational value.



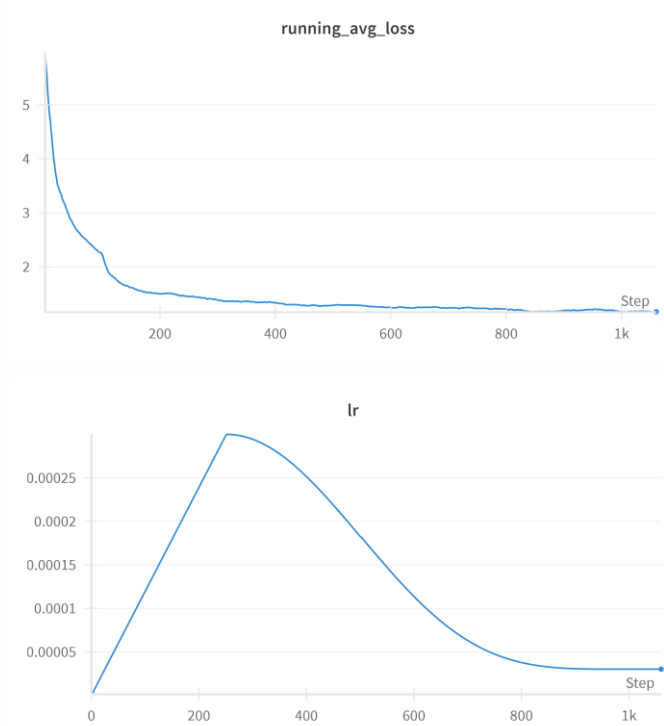
Hm...
Bxd5?



Methodology

- Tokenizer was initially trained on raw moves text, which turned out to be a bad, so I removed the move numbers to let the model be more efficient with its learning: "e4 e5 Nf3 Nc6 Bc4 ..."; this resulted in a vocabulary size of 530.
- I also tried to create my own transformer architecture but didn't get decent results, so I resorted to adapting Karpathy's NanoGPT, which performed much better.
- Trained two models (~25M vs ~100M parameters) until their loss curves plateaued at a mere 16K games.

	25M	100M
Val. Loss	1.16	1.49
Perp	3.19	4.45



Results

Game Length				
	25M vs Rand	25M vs Stfish	100M vs Rand	100M vs Stfish
Mean	9.5	8.1	7.7	7.6
Std	2.6	2.1	2.6	2.4
Min	5	4	3	3
Max	16	13	14	13

- It's surprising the smaller model was able to learn better; it's possible this simple architecture doesn't scale well for this task.
- The model seems to last longer vs random moves, something that I find very surprising seeing as the training data is vastly comprised of legitimate games.

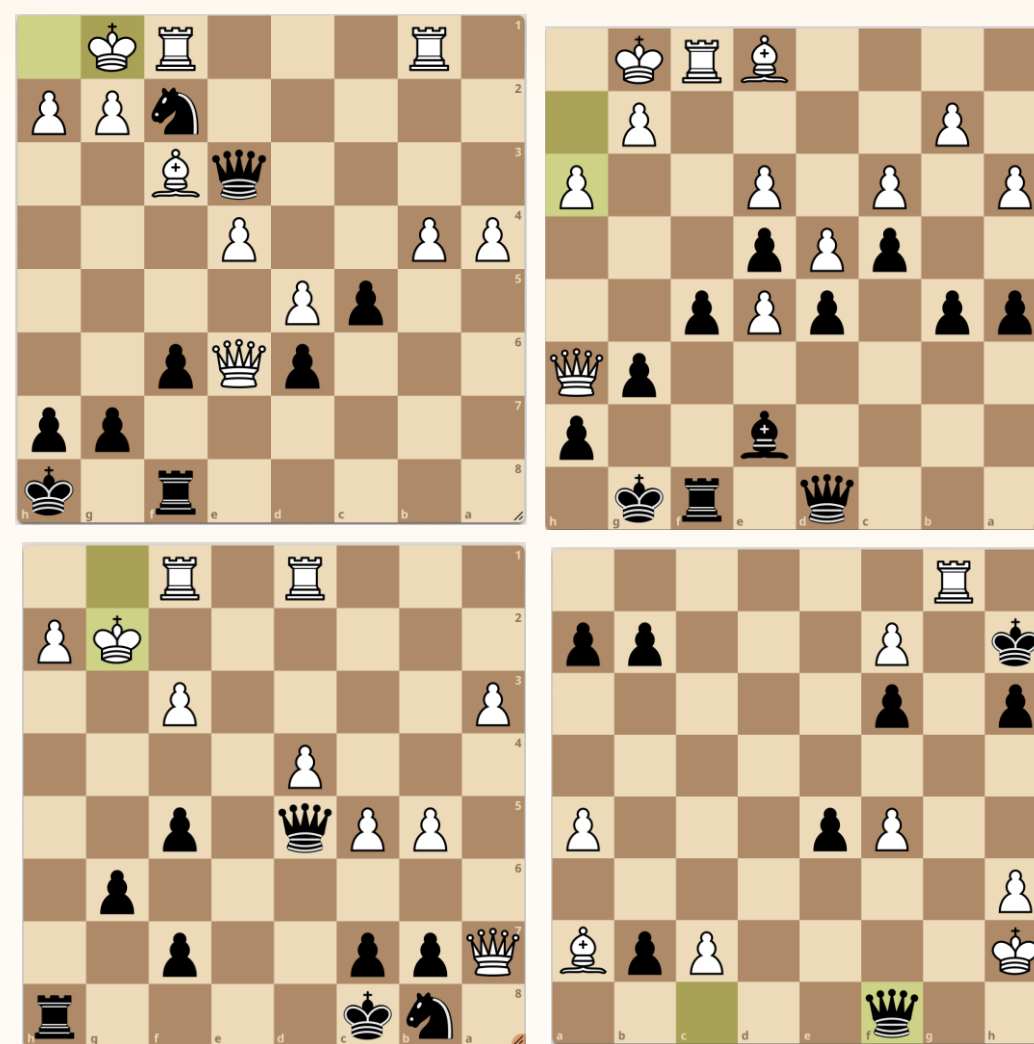
Checkmate Puzzle Identifier

Goal

To identify checkmate puzzles from non-checkmate puzzles using a CNN.

Dataset

The data used for this project came from the *Lichess Puzzles Database*. It consists of a FEN (Forsyth-Edwards notation) which represents the current position, and a series of moves along with the number of moves the puzzle takes to complete.



Can you identify which of these puzzles end in checkmate?

